Awesome title for your research paper

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Abstract

This is a document to showcase how github repos can be used, in combination with other things to create reproducible papers. Our aim is not to make a *real* paper, but to show how these tools can be used. Sometimes it takes some time to create these files, but ultimatelly they save us a lot of time, specially when we have to re-do analyses etc.

In this document I go through some stupid analyses but I hope the documents will be of use for you in the future ;)

1. Introduction

We all know this dataset known as iris. It is available directly from R and it has this and that variables.

This species are awesome and I could show you a picture here.

This introduction should be longer and more interesting... but it's not the point of this document.

2. Materials and Methods

2.1. Data

I will use the data directly from R, but I already wrote the csv file into de `./raw_data` folder. That is the raw data, but for some reason, we decided to remove 20 individuals randomly from the dataset, and save that into the `./data` folder.

That cleaning script can be called from inside the Rmarkdown file, whenever we need it (e.g. if we modify the script) adding `eval = TRUE`, or we can skip the chunk of code by setting `eval = FALSE`.

And if we are happy with our cleaning, we could just read in the cleaned dataset.

2.2. Analyses

I performed a bunch of stupid analyses in different scripts, so that, only those can be changed if needed.

Here you can include equations or formulas in the LaTeX format. For instance, you can inline equations: $response \sim predictor_1 * predictor_2$

Or

$$y_{ij} = b_{ij} + \beta_0 + \beta_1$$

3. Results

We did a PCA and here are the results are shown in Figure 1.

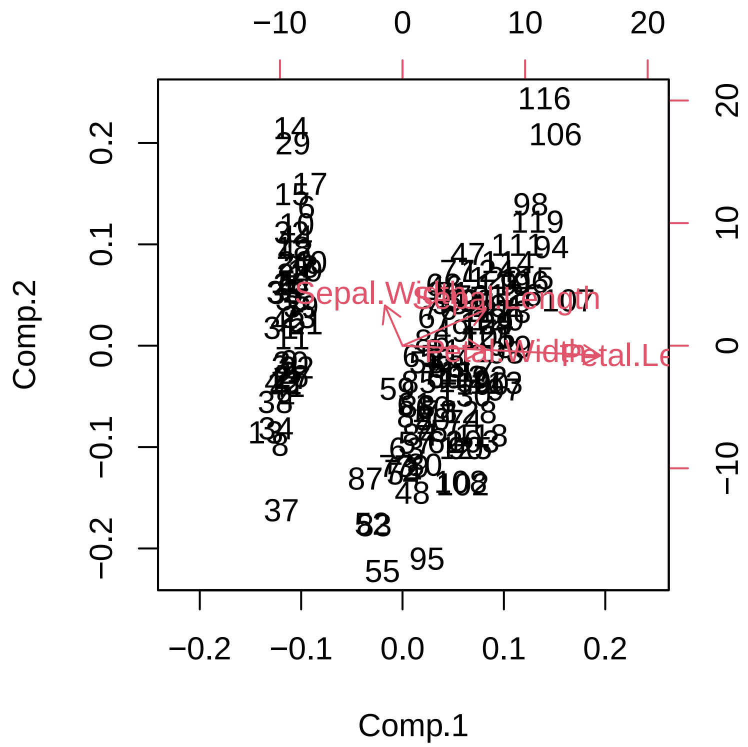

Figure 1: PCA plot.

The results of the ANOVA can be found in Table 1, testing for the relationship between this and that, using the species as a factor.

Table 1: ANOVA table for some stupid analysis.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Petal.Width	1	379.872712	379.8727120	3026.079684	0.0000000
Species	2	11.815674	5.9078371	47.062043	0.0000000
Petal.Width:Species	2	1.668297	0.8341485	6.644857	0.0018138
Residuals	124	15.566086	0.1255330	NA	NA

The results are awesome. I found that F(1; 124)=3026.08, $P\text{-}value = 5.7e\text{-}89$

4. Discussion

Write here the discussion of your paper. You can always add references like [Sutton and Higgins, 2008].

Write here the discussion of your paper.

```r
par(mfrow = c(1, 2))
with(iris, plot(Petal.Length ~ Petal.Width *
    Species))
```

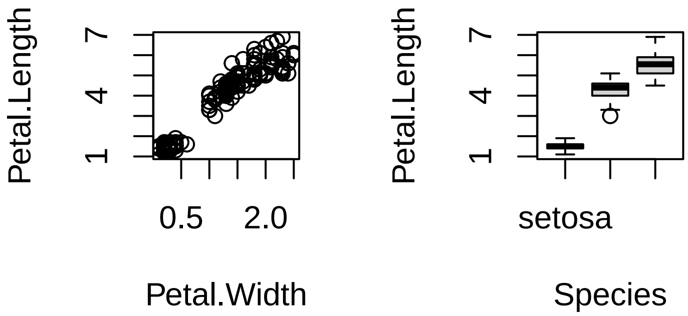

Figure 2: Another stupid plot.

Write here the discussion of your paper.

References

A. J. Sutton and J. P Higgins. Recent developments in meta-analysis. *Statistics in medicine*, 27(5):625–650, 2008.